

Exercise 1

Consider a robot with n joints that is redundant with respect to an end-effector velocity task $\dot{r} \in \mathbb{R}^m$, being $m < n$. At a given configuration q , the joint velocity command $\dot{q} \in \mathbb{R}^n$ adopts the Projected Gradient (PG) method, using a criterion $H(q)$ to be locally maximized, in the form

$$\dot{q} = \dot{q}_0 + J^\#(q)(\dot{r} - J(q)\dot{q}_0) \quad \text{with } \dot{q}_0 = \alpha \nabla_q H(q), \quad (1)$$

where $\alpha \in [0, \alpha_{\max}]$ is a scalar step to be chosen in the gradient direction of $H(q)$. The joint velocity is bounded componentwise by

$$-\dot{q}_{\max} \leq \dot{q} \leq \dot{q}_{\max} \iff -\dot{q}_{\max,i} \leq \dot{q}_i \leq \dot{q}_{\max,i} \quad i = 1, \dots, n. \quad (2)$$

Assuming that the minimum norm velocity solution $\dot{q}_r = J^\#(q)\dot{r}$ is feasible, i.e., satisfies all the bounds (2), determine the largest step α^* in its domain of definition so that the command (1) is still feasible.

Find the analytic expression of the optimal step α^* . Provide then a geometric illustration of the solution for the generic case of a robot with $n = 2$ joints that performs a task of dimension $m = 1$, representing the entire problem in the plane $(\dot{q}_1, \dot{q}_2)$.

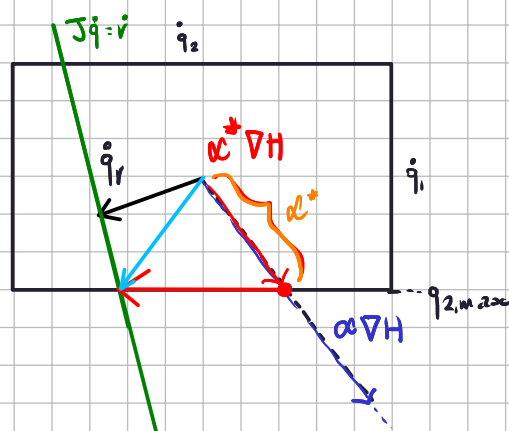
$$\dot{q} = \underbrace{\dot{q}_r}_{\dot{q}_r} + \underbrace{J^\#(q)(\dot{r} - J(q)\dot{q}_0)}_{\alpha \dot{q}_p} = \underbrace{J^\#(q)\dot{r}}_{\dot{q}_r} + \underbrace{(\mathbf{I} - J^\#(q)J(q))\dot{q}_0}_{\alpha \dot{q}_p} = (\mathbf{I} - J^\#(q)J(q))\alpha \nabla H + \alpha \underbrace{(\mathbf{I} - J^\#(q)J(q))\nabla H}_{\dot{q}_p}$$

≥ 0 due assumption

Assume $|\dot{q}_r| \leq \dot{q}_{\max}$, $\dot{q} = \dot{q}_r + \alpha \dot{q}_p$, $|\dot{q}| = |\dot{q}_r| + \alpha |\dot{q}_p| \leq \dot{q}_{\max} \Rightarrow \alpha |\dot{q}_p| \leq \dot{q}_{\max} - |\dot{q}_r|$

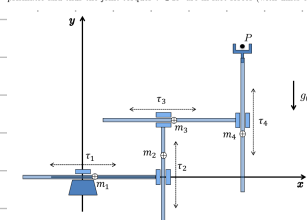
$\Rightarrow \alpha \leq \frac{\dot{q}_{\max} - |J^\#(q)\dot{r}|}{|(\mathbf{I} - J^\#(q)J(q))\dot{q}_0|}$ This is a vector of n values for α , hence we have to choose the smallest.

$\alpha^* = \min_i \frac{\dot{q}_{\max,i} - |J^\#(q)\dot{r} e_i|}{|(\mathbf{I} - J^\#(q)J(q))\dot{q}_0 e_i|}$ where $e_i = i$ -th col of Identity $e_i = (0 \dots 1 \dots 0)^T$



Exercise 2

Consider the planar 4-dof robot shown in Fig. 1, moving under gravity. All robot joints are prismatic and thus the joint torques $\tau \in \mathbb{R}^4$ are in fact forces (with units expressed in [N]).



- Derive in symbolic form the dynamic model of this robot in the joint space.
- Determine the 2×2 Cartesian inertia matrix M_r at the end-effector level. Compute then M_r numerically for $q = 0$ and equal link masses $m_i = 1$ [kg], $i = 1, \dots, 4$.
- Suppose you have no knowledge about the values of the dynamic parameters of this 4P robot (except for the gravity acceleration $g_0 = 9.81$ m/s²). Design an adaptive control law for τ that is able to globally asymptotically stabilize the tracking error along a smooth desired trajectory $q_d(t)$. The controller should be of minimal dimension.

let d_i to be the distance between CoM_i and RF_i

$$T = \frac{1}{2} [m_1 \dot{q}_1^2 + m_2 (\dot{q}_1^2 + \dot{q}_2^2) + m_3 (\dot{q}_1^2 + \dot{q}_2^2 + \dot{q}_3^2 + 2\dot{q}_1\dot{q}_3) + m_4 (\dot{q}_1^2 + \dot{q}_2^2 + \dot{q}_3^2 + 2\dot{q}_1\dot{q}_3 + \dot{q}_4^2 + 2\dot{q}_2\dot{q}_4)]$$

$$U = g_0 (m_2 (q_2 - d_2) + m_3 q_2 + m_4 (q_2 + q_4 - d_4))$$

$$g = (0 \quad m_2 + m_3 + m_4 \quad 0 \quad m_4)^T = (0 \quad e_3 \quad 0 \quad e_4)^T$$

$$M_{11} = m_1 + m_2 + m_3 + m_4 \quad M_{13} = m_3 + m_4 \quad M_{22} = m_2 + m_3 + m_4 \quad M_{24} = m_4 \quad M_{33} = m_3 + m_4 \quad M_{44} = m_4$$

$e_1 \quad e_2 \quad e_3 \quad e_4$

$$M = \begin{pmatrix} e_1 & 0 & e_2 & 0 \\ 0 & e_3 & 0 & e_4 \\ e_2 & 0 & e_2 & 0 \\ 0 & e_4 & 0 & e_4 \end{pmatrix} \quad \text{the Jacobian} \quad J = \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix}$$

the inertia matrix is $M_r = (J \bar{M}^{-1} J^T)^{-1} = \begin{pmatrix} e_2 & 0 \\ 0 & e_4 \end{pmatrix}$

Now i linearly parametrize:

$$\begin{cases} e_1 \ddot{q}_1 + e_2 \ddot{q}_3 = \tau_1 \\ e_3 \ddot{q}_2 + e_4 \ddot{q}_4 + e_3 = \tau_2 \\ e_2 \ddot{q}_1 + e_2 \ddot{q}_3 = \tau_3 \\ e_4 \ddot{q}_2 + e_4 \ddot{q}_4 + e_4 = \tau_4 \end{cases} \Rightarrow \begin{bmatrix} \ddot{q}_1 & \ddot{q}_2 & 0 & 0 \\ 0 & 0 & \ddot{q}_2 + 1 & \ddot{q}_4 \\ 0 & \ddot{q}_1 + \ddot{q}_3 & 0 & 0 \\ 0 & 0 & 0 & \ddot{q}_2 + \ddot{q}_4 + 1 \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \\ e_3 \\ e_4 \end{bmatrix} = \tau$$

Y have linearly independent rows hence the minimal number of dynamic coefficients is 4.

Consider $\Gamma, K_P, K_D, \Lambda$ Four diagonal 4×4 positive definite matrices.

let $\dot{q}_r = \dot{q}_d + \Lambda(q_d - q)$ so Y_r

$$\tau = \begin{bmatrix} \ddot{q}_{1r} & \ddot{q}_{2r} & 0 & 0 \\ 0 & 0 & \ddot{q}_{2r} + 1 & \ddot{q}_{4r} \\ 0 & \ddot{q}_{1r} + \ddot{q}_{3r} & 0 & 0 \\ 0 & 0 & 0 & \ddot{q}_{2r} + \ddot{q}_{4r} + 1 \end{bmatrix} \begin{bmatrix} \hat{e}_1 \\ \hat{e}_2 \\ \hat{e}_3 \\ \hat{e}_4 \end{bmatrix} + K_P(q_d - q) + K_D(\dot{q}_d - \dot{q}), \quad \hat{e} = \Gamma Y_r^T (\dot{q}_r - \dot{q})$$

Exercise 3

a) $\ddot{r} = J_t \ddot{q} + \dot{J}_t \dot{q} \Rightarrow \ddot{q} = J_t^\# (\ddot{r} - \dot{J}_t \dot{q})$

$$\ddot{q} = \bar{M}^{-1} (\tau + J_g^T F - c) \Rightarrow \ddot{r} = J_t \bar{M}^{-1} (\tau + J_g^T F - c) + \dot{J}_t \dot{q} \Rightarrow$$

$J_t \bar{M}^{-1} \tau = \ddot{r} - J_t \bar{M}^{-1} (J_g^T F - c) - \dot{J}_t \dot{q}$ i have to consider the weighted pseudo inverse of $J_t \bar{M}^{-1}$ weighted by $\bar{M}^{-1}$.

$$(J_t \bar{M}^{-1})_{\bar{M}^{-1}}^\# = \bar{M} \bar{M}^{-1} J_t^T (J_t \bar{M}^{-1} \bar{M} \bar{M}^{-1} J_t^T)^{-1} = J_t^T (J_t \bar{M}^{-1} J_t^T)^{-1}$$

c) For a PR:

$$M = \begin{bmatrix} 2 & -\frac{1}{4} s_2 \\ -\frac{1}{4} s_2 & \frac{13}{80} \end{bmatrix} \quad c = \begin{cases} -\frac{1}{4} c_2 \dot{q}_2^2 \\ 0 \end{cases} \quad \text{in the given state } M = \begin{pmatrix} 2 & -0.25 \\ -0.25 & 0.16 \end{pmatrix} \quad c = 0$$

$$J_t = (1 \quad -l_2 s_2) = (1 \quad -0.5) \quad J_g = \begin{pmatrix} 1 & -l_2 s_2 \\ 0 & l_2 c_2 \end{pmatrix} = \begin{pmatrix} 1 & -0.5 \\ 0 & 0 \end{pmatrix} \quad \ddot{r}(0) = 9 \cos(3 \cdot 0) = 9$$

$$(J_t \bar{M}^{-1})_{\bar{M}^{-1}}^\# = J_t^T (J_t \bar{M}^{-1} J_t^T)^{-1} = \begin{pmatrix} 0.6363 \\ -0.3181 \end{pmatrix}$$

$$\ddot{r} - J_t \bar{M}^{-1} (J_g^T F - c) = 10.57 \Rightarrow \tau = \begin{cases} 6.725 \\ -3.3623 \end{cases}$$

b) We add to sum to the desired $\ddot{r}$ a PD controller

$$z = J_t^T (J_t \bar{M}^{-1} J_t^T)^{-1} (\ddot{r}_d - J_t \bar{M}^{-1} (J_t^T F - c) - \dot{J}_t \dot{q}) \text{ becomes:}$$

$$z = J_t^T (J_t \bar{M}^{-1} J_t^T)^{-1} (\ddot{r}_d + \underset{\uparrow r_d - r}{K_p e} + \underset{\uparrow \dot{r}_d - \dot{r}}{K_d \dot{e}} - J_t \bar{M}^{-1} (J_t^T F - c) - \dot{J}_t \dot{q})$$

Exercise 4

Consider the same PR robot of Exercise 3c), but now moving freely in a vertical plane, without any interaction with the environment. Design a FDI system that enables to detect and isolate any fault of the robot actuation system, without the need to measure the joint torques, nor compute accelerations, nor invert the inertia matrix. Provide the explicit and computable expression of the so-called residual vector $r \in \mathbb{R}^2$. Sketch the behavior of the two residual components when a total fault of the first actuator occurs at an instant t when the robot is in the state $q(t) = (0, \pi/2)$ [m, rad], $\dot{q}(t) = (1, -1)$ [m/s, rad/s], and a constant torque $\tau = (5, -2)$ [N, Nm] is being requested by the control law.

$$M = \begin{bmatrix} 2 & -\frac{1}{4} s_2 \\ -\frac{1}{4} s_2 & \frac{13}{80} \end{bmatrix}$$

We choose a diagonal pos. definite matrix K , we need to compute the

$$\alpha_i = -\frac{1}{2} \dot{q}^T \frac{\partial M}{\partial q_i} \dot{q} + g_i \quad p_{c2} = l_2 s_2 = \frac{1}{2} s_2$$

$$\text{First of all: } g = \begin{pmatrix} 0 & \frac{1}{2} g_0 s_2 \end{pmatrix}^T \Rightarrow \alpha_1 = 0$$

$$\frac{\partial M}{\partial q_1} = 0 \quad \frac{\partial M}{\partial q_2} = \begin{pmatrix} 0 & -\frac{1}{4} c_2 \\ -\frac{1}{4} c_2 & 0 \end{pmatrix} \quad \alpha_2 = \frac{1}{2} g_0 s_2 + \frac{1}{4} c_2 \dot{q}_1 \dot{q}_2$$

$$r = \begin{cases} \int \kappa_1 (\tau_1 - r_1) dt - 2\dot{q}_1 + \frac{1}{4} s_2 \dot{q}_2 \\ \int \kappa_2 (\tau_2 - \frac{1}{2} g_0 s_2 - \frac{1}{4} c_2 \dot{q}_1 \dot{q}_2 - r_2) dt + \frac{1}{4} s_2 \dot{q}_1 - \frac{13}{80} \dot{q}_2 \end{cases} \Rightarrow \dot{r} = K (z_s - r)$$

If a total fault occurs $\tau_{s1} = -5$ so

$$\dot{r}_1 = \kappa_1 (-5 - r_1) \quad \text{we choose } \kappa_1 = 2$$

$$\dot{r}_1 = -10 - 2r_1 \Rightarrow r_1 s + 2r_1 = -10 \Rightarrow r_1(s+2) = -10 \Rightarrow r_1(s) = -\frac{10}{s+2} \Rightarrow \lim_{t \rightarrow \infty} r_1 = -5 = \tau_{s1}$$

The profile of r_1 is exponential.